Fiscal Year:	FY 2006	**Task Last Updated:**	FY 04/13/2006
PI Name:	Yu, Yongjia Ph.D.		
Project Title:	Impact of HZE Particles Exposure on Proliferation and Differentiation of Human Neural System Cells		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	(1) **BMed**:Risk of Adverse Cognitive or Behavioral Conditions and Psychiatric Disorders		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	77555-0884	**Congressional District:**	9
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2003 Space Radiation Biology and Shielding 03-OBPR-07
Start Date:	05/15/2004	**End Date:**	06/30/2008
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	End date changed to 6/30/2008 per JSC info update (10/06)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Chuang, Eric (National Institutes of Health) Morrill, Steve (UTMB) wu, Ping (UTMB)		
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Task Description:	Space radiations such as HZE particles pose a great risk of damaging the central nerve system (CNS) of astronauts. Human neural stem cells (hNSCs) are of particular interest since functional NSCs in adult brain are potential targets of HZE damage, and because exogenous hNSCs may provide potential therapeutic source for replacing lost neurons in radiation-damaged brain. The long-term objective of this project is to study the mechanisms underlying HZE particles-induced adverse impact on the proliferation and differentiation of a primary human fetal neural stem cell line, K048. K048 retains self-renewal and multipotential differentiation capabilities in vitro and in vivo. Moreover, a novel in vitro priming procedure appears to direct K048 cells to adopt neuronal differentiation fate. We will test two hypotheses:(1) HZE particles disrupt proliferation and differentiation of human NSCs by activating apoptotic and genotoxic-response pathways. (2) Inhibitory neurogenic micro-environmental cues produced by HZE in rat brain can be overcome by this novel in vitro priming prior to transplantation into brain. We will pursue three specific aims: (1) Determine the impact of 56Fe and gamma-ray irradiation on proliferation and differentiation of K048 cells in vitro. (2) Determine gene expression patterns and apoptotic responses of proliferating and differentiating K048 cells following 56Fe and gamma-ray irradiation. (3) Determine whether in vitro primed K048 cells can differentiate into neurons when grafted into neurogenic and non-neurogenic regions of rat brains damaged by 56Fe and gamma-rays. The results of this project will contribute to a better understanding of cellular and molecular mechanisms of damage to the CNS following irradiation by HZE particles. To determine the impact of HZE on proliferation and differentiation of K048 neural stem cells in vitro. To determine the fate of primed K048 cells grafted into 56Fe-irradiated rat brains. In this project, we propose to examine the impact of HZE and low LET radiations on the proliferation and differentiation of K048 human neural stem cell line. The K048 cell model is a unique experimental system not only because of its human origin, but also because this line is not transformed and retains the characteristics of normal neural stem cells. Using this system we will address directly to one of the high priority topics listed in NRA 03-OBPR-07 Research Announcement, namely “What are the cellular and molecular mechanisms of damage to the CNS following irradiation by HZE particles?”. The proposed research is also within the scope of “Critical Path Roadmap”, regarding risk number (39) “Degenerative tissue diseases including damage to the central nervous system, cataracts, circulatory diseases, or other late non-cancer risks”, and risk number (41) “Early or acute effects from radiation exposure”. Specifically, our proposed studies address Critical Question (10.05) in radiation, “Are there unique biological effects associated with HZEs?”
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	In this reporting period, we have conducted further experiments based on our previous studies to determine the does-response and timing-response relations in low LET gamma-rays induced inhibition of neuronal differentiation. We found that: (1)gamma-irradiation inhibited the neuron production in K048 differentiation in vitro, but had little effect on astroglia generation. Approximately 40% of differentiated population from unirradiated K048 cells stained TuJ1 positive, a single dose of 0.5 Gy irradiation reduced this figure to about 20%. Following a singe 2 Gy irradiation, only about 2% of surviving stem cells differentiated into neurons. In contrast, approximately 30% of unirradiated stem cell population differentiated into astroglia, this percentage remained similar even in population exposed to 2.0 Gy of gamma-rays. (2)when K048 cells were irradiated with 1 Gy gamma-rays either 1 day prior to, or 1,3,7 days post-initiation of differentiation , the percentages of new neurons were all reduced in comparison to untreated control. However, the inhibitory effect of gamma-rays on neuronal differentiation appears to be more prevelent during early stages of differentiation . In addition, we had our first run at the NSRL at BNL, in which K048 cells were exposed to 1000MeV/n 56Fe ion radiation. While complete data from this run is still being collected, a preliminary result clearly indicates a dose-dependent inhibition of neuronal differentiation of K048 cells by HZE particles.
Bibliography Type:	Description: (Last Updated: 03/19/2010)
Abstracts for Journals and Proceedings	Yu Y, Cittelly D, Wu P, Cornforth MN. "Inhibition of Human Neural Stem Cell Differentiation by gamma-Rays." Presented at Bioastronautics Investigators' Workshop, Galveston, TX, January 10–12, 2005. Abstracts, Bioastronautics Investigators' Workshop, Galveston, TX, January 10–12, 2005. Online Pub: http://www.dsls.usra.edu/meetings/bio2005/pdf/SRPosters/SRPosters.pdf , Jan-2005
Abstracts for Journals and Proceedings	Yu Y, Gao Y, Cornforth MN, Morrill SM, Gao JL,Cittelly D, Wu P. "Response of Human Neural Stem Cells to Ionizing Radiation." Presented at 16th Annual NASA Space Radiation Investigators' Workshop, Port Jefferson, NY, May 15-18, 2005. Abstracts, 16th Annual NASA Space Radiation Investigators' Workshop, Port Jefferson, NY, May 15-18, 2005. http://www.dsls.usra.edu/meetings/radiation2005/pdf/1038yu.pdf , May-2005
Abstracts for Journals and Proceedings	Yu Y, Gao Y, Cornforth MN, Morrill SM, Gao JL, Wu P. "Differentiation of grafted human neural stem cells in irradiated rat brain. " Presented at the Annual Meeting of Radiation Research Society, Denver, Co., October 16-19, 2005. Abstracts, Annual Meeting of Radiation Research Society, Denver, Co., October 16-19, 2005. , Oct-2005